Reflection on the interactive Trading Session #4

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Abstract

This note covers the 4th trading session, where I was an asset exchanger in a two-stock market in which only dealers could quote. Part 1 covers my task, my approach and 4 key moments that decided my result. Part 2 uses the official session data to study trading costs, order choice and order aggressiveness, with the paper Bertsimas and Lo (1998) as a benchmark. I traded with a teammate, but I write "I" throughout for simplicity; also the quick clicks behind the biggest mistakes and gains were mine, since I had the mouse.

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Part 1: My Session 4

1.1 Setup and task

The session lasted 60 minutes and opened with empty books in both stocks. It built on the earlier sessions: the deadline of Session 1, where bottles still held at the bell were worth nothing; the news traders of Session 2, who knew the fundamental value; the two stocks and the asset exchanger role of Session 3; and the lesson of the open outcry that trading first means paying for price discovery. Table 1 lists the roles.

Table 1. Roles in Session 4 (course briefing and participant file).

Role	Seats	Orders	Information	Objective
Dealers	28	Limit and market	None	Earn the spread, ±20,000 per stock
News traders	14	Market only	Fundamental value	P&L marked to value
Asset exchangers	14	Market only	None	End flat, ±5,000 band

I had to sell 10,000 STOCK1 and buy 10,000 STOCK2 within the hour and stay within 5,000 shares of a straight line increasing/decreasing (depending on the stock) from my starting position to zero. Exchangers were scored within their group on a VWAP measure, the band and the final target, with equal weights. With market orders only and no information on value, I paid the spread on every trade, and my only decisions were when and how much to trade.

1.2 Expectations and approach

- I expected dealers to quote tight spreads from the start, as the order book had been liquid in Session 3; instead the open was chaotic.
- With market orders only, I expected the spread to be my main cost and thought I could keep it small by being patient.
- I expected prices to stay close to value, since dealers had no information and news traders would correct mispricing.

As lesson 1 says, I should have decided an execution rule before the bell. Instead, I only planned to trade when the spread was small and to keep orders small, so as to average my buying and selling prices. In practice I used orders of 500 to 1,000 shares until my last STOCK2 purchases (2,000 to 3,000), and traded rarely: 28 orders and 43 fills. In fact, only 3 of the 46 active traders in the classroom had fewer fills than I.



Figure 1. Best bid and ask against fundamental value, median per 30 seconds.

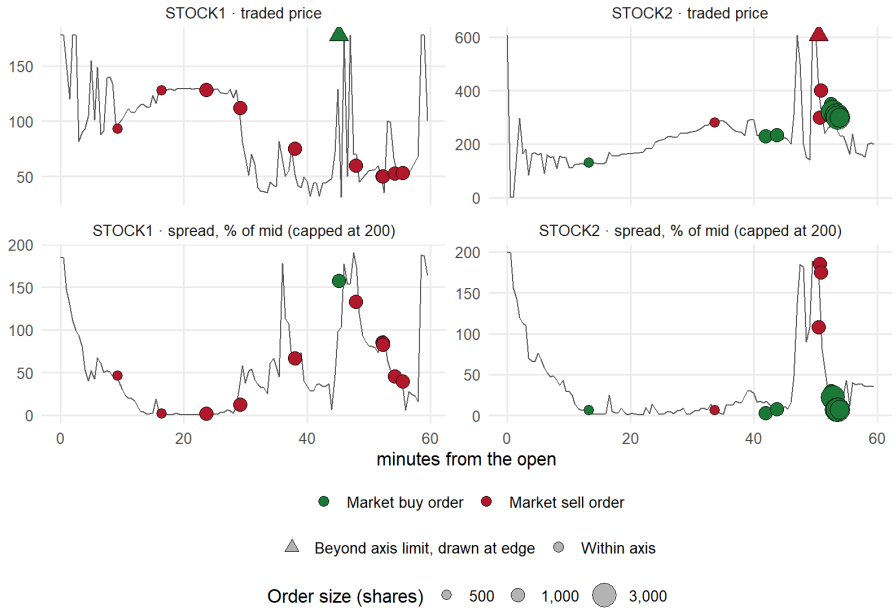


Figure 2. My market orders against the traded price and the quoted spread as % of the midquote (dot size: shares).



Figure 3. My position against the ±5,000 band. STOCK2 left the band at minute 33.

1.3 Four key moments

Figure 1 shows the market, Figure 2 my orders and Figure 3 my position against the band.

What can we get out of the three figures: urgency built up!

STOCK1. The spread was tight between minutes 10 and 30 (Figure 1), and that is when I made my best sales. After minute 30 the book widened and I sold only 1,000 more shares until minute 50, when I still held 6,000 against about 1,700 on the line (Figure 3). I stayed inside the band, but the last 6,000 went at the deadline, at less than half my earlier prices (moment 4).

STOCK2. The spread remained relatively tight until around minute 40, yet I barely traded. My position therefore fell behind the required trajectory and moved outside the ±5,000 corridor around minute 33 (Figure 3). By delaying when liquidity was available, I turned a flexible execution problem into an urgent one: later in the session, inventory management rather than the spread increasingly determined when I had to trade.

1. The wild opening of STOCK1 (minutes 0 to 9). In the first three minutes STOCK1 printed between 12 and 9,999.99, and its spread stayed above 60% of the midquote for ten minutes, so I stayed out.

2. Leaving STOCK2 behind, and the big error (minutes 13 to 44). STOCK2 was the most liquid book, yet I bought only 500 shares by minute 40. I also made a huge mistake by selling 500 at 281 at minute 33.7, against my task. It was a confusion as I spent too much time looking at the STOCK1 screen, then took my eyes away from the computer and then looked back but at STOCK2 this time and acted on it with my brain still wired to STOCK1's trading goal (had to sell so was looking at bids, not asks).

3. The fat finger (minute 50.5). A dealer bid 3,000 for STOCK2, worth 173.8. I sold 1,000 into that bid, and buying them back near 300 left me about 2.7 million ahead. It felt like the market paying me back for the mix-up five minutes earlier.

4. The deadline (minutes 52 to 56). In 90 seconds I bought 13,000 STOCK2 at 300 to 350, about twice its value, and sold my last 6,000 STOCK1 at 50 to 53, less than half my earlier prices.

1.4 Did I succeed?

I completed the task but executed it badly. In my group I sold STOCK1 at the lowest average price and bought STOCK2 at the highest, and ranked fifth of the five scored exchangers (Table 2).

Table 2. Official results for exchangers who sold STOCK1 and bought STOCK2 (T4-results.pdf).

Trader	Score	P&L (000)	Sell price STOCK1	Buy price STOCK2	Done
Trader A	0.56	−1,498	83.62	243.53	No
Trader B	0.43	−1,355	106.67	241.46	No
Trader C	0.36	−803	105.94	197.66	Yes
Trader D	0.16	−1,282	84.18	211.00	Yes
trader48 (me)	−0.13	+341	72.21	297.34	Yes

My P&L of +341,026 was the only positive one in the group. Yet, without the fat finger (moment 3) I would have lost about 2.4 million, the worst result of the five. I had no rule for when to trade, I left the liquid window of minutes 10 to 40 unused for STOCK2, I confused the books under pressure, and I let the deadline set my prices. Very bad overall…

1.5 Lessons

The 3 lessons I would remember are:

1. Decide the execution rule before the bell, around the band's line. In this session liquidity was better early than late, and the line, which has two thirds of the task done by minute 40, would have put most of my trading in the tight-spread window of minutes 10 to 40. Following it at a steady pace while skipping spreads wider than 20% would have bought STOCK2 at 212 instead of 297 in the replay of Section 2.4.
2. Check the stock and side before every order. With opposite objectives in the two stocks, I should have kept a simple “SELL STOCK1 / BUY STOCK2” rule visible beside the screen.
3. Judge execution against a benchmark. My only profit came from another trader's mistake and being aware of it is the first step toward improvement.

Part 2: Trading costs, order choice and order aggressiveness

2.1 Question

A trader with a deadline in a dealer market pays for immediacy on every order. This part measures how large that cost was across the market, when it was high, and whether a simple execution rule could have reduced it. To answer it, I first take an explanation from outside the data, the execution model of Bertsimas and Lo (1998), which says how a trader with a deadline should spread orders over time (2.2). I then describe liquidity during the hour from the quotes, and check that it persisted, as the model needs (2.3). Section 2.4 uses two implications of the model as benchmarks for interpreting the session data; and Section 2.5 lists what the data show that the model cannot explain.

The unit of analysis is the market: all quotes, and the orders of all 14 exchangers across both tasks.

2.2 Bertsimas and Lo (1998)

Bertsimas and Lo (1998) study how to execute a fixed quantity before a deadline at minimum expected cost. Under simple assumptions, the optimal strategy divides the order evenly over time. When persistent information about market conditions is available, execution can be adjusted away from this baseline. The framework gives two benchmarks for the session: keep a regular execution pace, and deviate from it temporarily when market conditions are unusually poor.

Session 4 differs slightly from the model, but the comparison is still useful. Asset exchangers faced a fixed deadline, could observe quoted spreads, and could only use market orders, so their main choices were when to trade and how much. The quoted spread is a simple, observable measure of current liquidity conditions. If these conditions persist, a short delay may improve execution; if they do not, delaying only increases the quantity that must be traded later. The analysis therefore asks two questions: did liquidity conditions persist during the session, and was exchanger execution concentrated in periods of relatively tight or wide spreads?

2.3 Liquidity during the hour

Figure 4A splits the hour into three phases: spreads were wide at the open (minutes 0 to 10), tight in the middle (to minute 30 in STOCK1 and minute 40 in STOCK2, below 20% about 90% of the time), and wide again at the end, when dealers obliged to quote often posted prices far from value. These phases lasted (Figure 4B): when the spread was tight, it was still tight one minute later 81 to 89% of the time; when it was wide, only 10 to 16% of the time.

The liquidity state therefore showed substantial short-run persistence: a trader observing a relatively tight spread had some reason to expect favourable conditions to persist. This makes short delays potentially useful, but does not imply that waiting indefinitely is optimal.

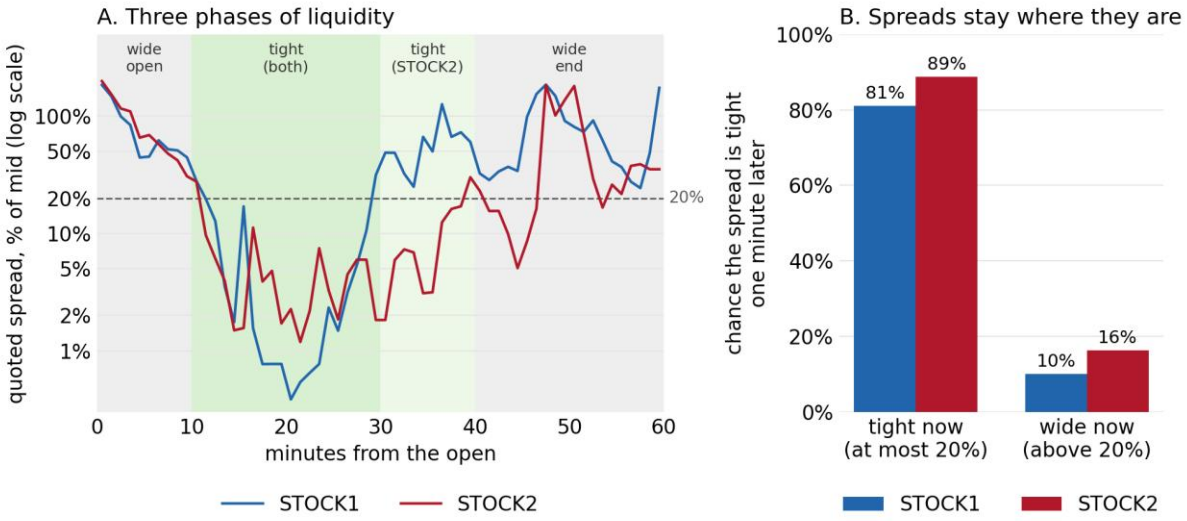


Figure 4. A: median quoted spread per minute, with the three phases shaded. B: chance that the spread is at most 20% one minute later, when it is tight or wide now.

2.4 Evidence from the session data

Each test takes one prediction of the model from Bertsimas and Lo (Section 2.2) and checks it against the session data: the official quotes, orders, trades and fundamental values.

A. Trading costs.

Prediction: adapting an even split to market conditions lowers the expected cost. Trading costs (also called execution or transaction costs) are here the spread and price impact paid (slippage), since the session had no fees. So the rule with the better average price had the lower cost.

I backtested two rules on the official quotes.

- the even split, which corresponds to the band's straight line: 10 orders of 1,000 shares at the best quote, at minutes 3, 9, ..., 57
- the adaptive rule, keeps those due times but when ask - bid exceeds a fifth of the midquote it waits for the first second when it no longer does

Each rule's price is the average of its 10 fills. Because wide spreads persisted (Figure 4B), waiting skips the worst quotes, such as an ask of 1,000 at minute 51 for a stock worth about 170. The replay covers all four exchanger legs (Figures 5 and 6), and I compare it with the median exchanger on each leg, because a few fills at stub quotes pull the averages far from typical prices. Figure 5 shows how the two rules differ: the adaptive rule stops in the grey zones and catches up once spreads tighten. Figure 6 shows what that difference is worth in price on each leg, next to every active exchanger. Waiting out wide spreads sold STOCK1 at 93.7 (even split 75.9, median exchanger 84.2), bought STOCK2 at 212.3 (290.6 and 241.5) and bought STOCK1 at 102.1 (119.9 and 114.5). The exception is selling STOCK2, where it got 187.8 against 191.3 and 199.0. That fits Bertsimas and Lo (pp. 15 and 16): adapting lowers the expected cost, not every realised one.

Weaknesses: it is a single replay with fills assumed at the displayed quote, although sometimes only 50 to 100 shares were shown. Also, it suffers from the biggest flaw of back testing: the market cannot react to the replayed orders, so there is no way to know what would have happened if they had really been sent.

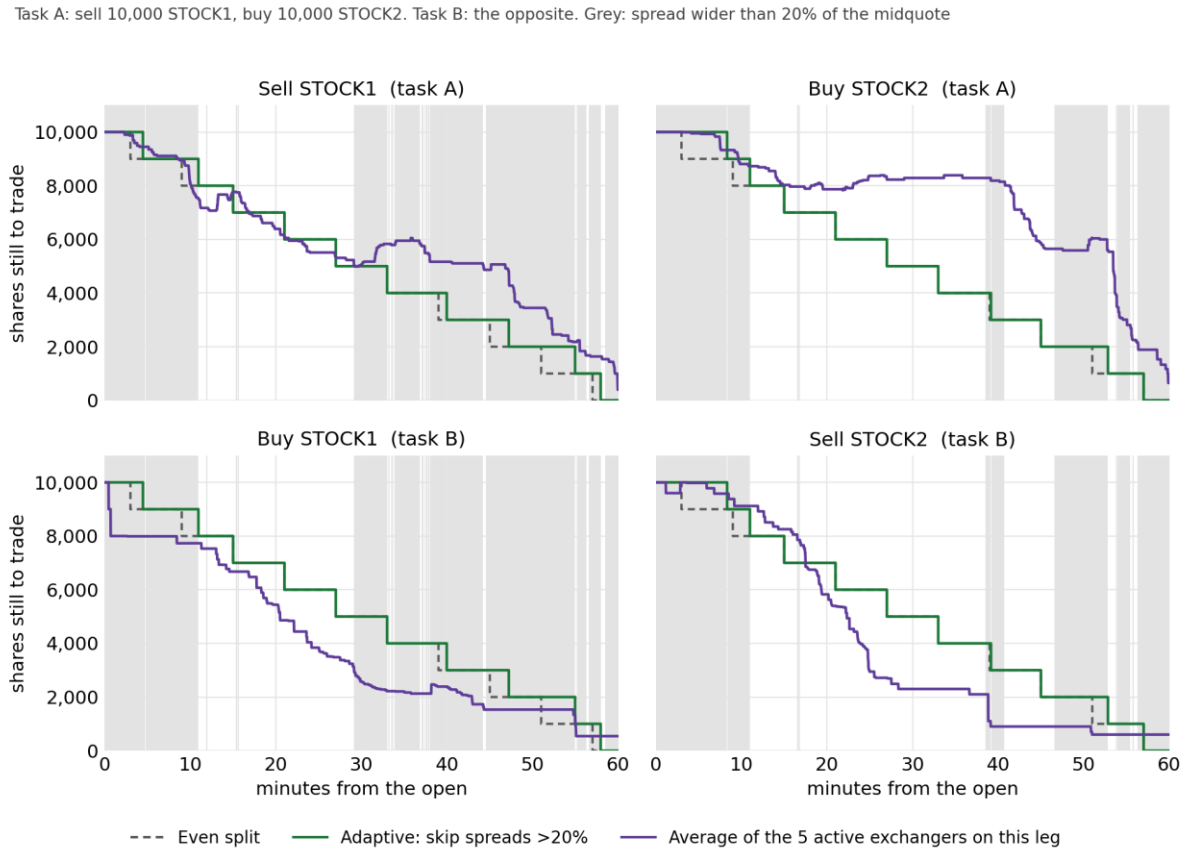


Figure 5. Shares left to trade on each of the four exchanger legs. Grey zones are when spread is above 20%.

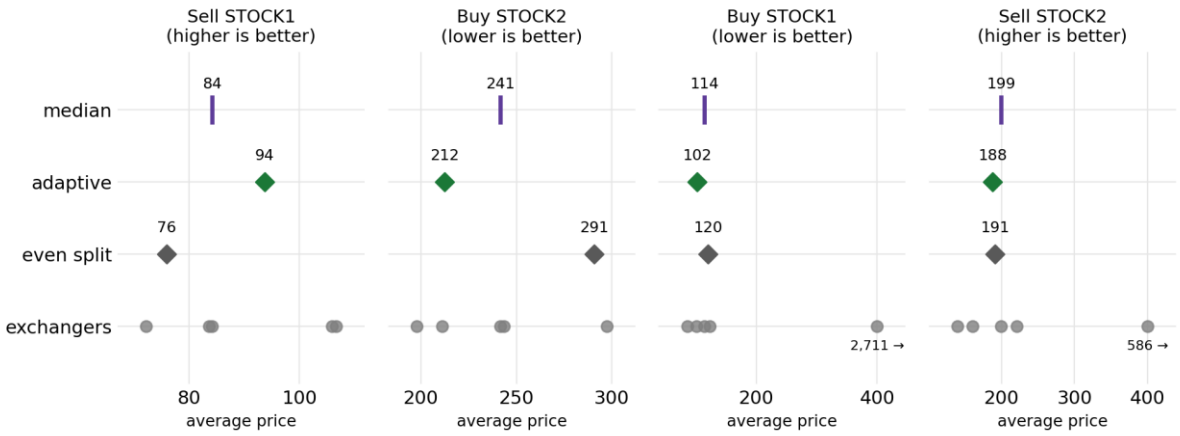


Figure 6. Average prices on each leg: replayed strategies, the median exchanger and each active exchanger.

B. Order choice and aggressiveness.

Prediction: because spreads persisted, a trader should trade while they are tight rather than cross wide ones. With market orders only, exchangers had two choices left: when to trade, and so which spread to accept. Figure 7 shows both for the whole market: each bar is the volume exchangers traded in the direction of their task during ten minutes, coloured by the spread each order crossed (blue below 20%, orange and red above), and the green zones mark the tight window of Section 2.3. Exchangers did use that window: between minutes 10 and 30, 39% of their volume traded, almost all of it at spreads below 20%. But they also traded at the open and in the last 20 minutes at whatever spread was quoted, so about half of their volume crossed a spread above 20%, and waiting was costly: among the five exchangers who had to buy STOCK2, the three who left at least 90% of it to the last 20 minutes paid 241 to 297 a share, against 198 and 211 for the two who bought about half earlier. The model's advice held in the middle of the hour, when trading cheaply was easy, and was ignored at the deadline, when most exchangers still had shares left and took the spread of the moment.

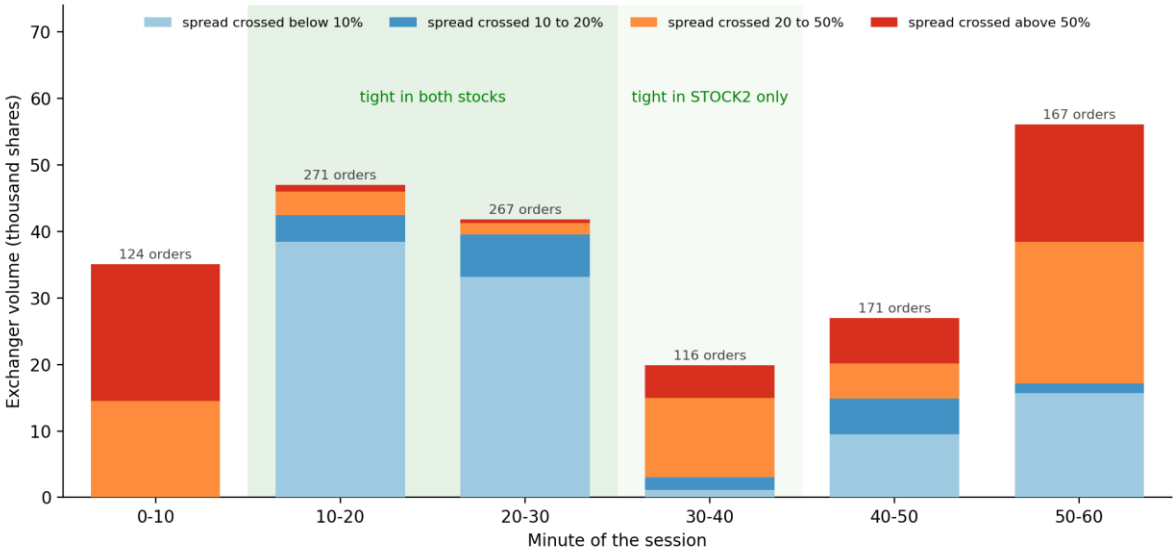


Figure 7. Volume traded by all exchangers in the direction of their task, per ten minutes, split by the spread each order crossed. Green: tight window (minutes 10 to 30 in both stocks, to 40 in STOCK2).

2.5 What the model leaves out

Section 2.4 used the paper to explain what the data show; this section shows what the paper leaves unexplained in the data.

First, the paper does explain well the main pattern in the data: a steady pace is the right baseline, and waiting out wide spreads pays because they last.

Yet, two things in the data fall outside it.

- Stub quotes (placeholder prices far from value that dealers posted to meet their obligation to quote) and a few fat-fingered bids (STOCK2 bids of 2,000 and 3,000 around minute 50, for a stock worth about 174, hit for 17,000 shares; STOCK1 bids of 110 to 190 between minutes 38 and 49, worth 33 to 45, hit for 7,000) made the cost of an order jump instead of rising smoothly with its size
- All exchangers facing the same deadline, crowded into the same last minutes (Figure 7), which a model of a single trader cannot capture.

Across the market, one exchanger bought 12,000 STOCK1 at an average of 2,711 a share, far above value, another bought 92,170 STOCK1 in the last two minutes although its task was to sell

With one session and 11 active exchangers, this is a case study: the paper explains the broad pattern of the hour, but not the few extreme trades that decided several exchangers' results.

References

Bertsimas, D. and Lo, A. W. (1998). Optimal control of execution costs. Journal of Financial Markets

FIN11 course materials (2026)

Use of AI tools

Claude Opus 5-5 was used to write the code for the graphs, and start drafting.

Then, ChatGPT 6 Astra was used to double check data related work and improve overall formatting of the paper.